MAY 2018 FINAL EXAM SOLUTION (SAMPLE SOLUTION)

1(a). $n = 20$, $\sum_{i=1}^{20} x_i = 146$, $\sum_{i=1}^{20} x_i^2 = 1484.76$, $\sum_{i=1}^{20} (x_i - \bar{x})^3 = 2511.876$

Sample Mean, $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{\sum_{i=1}^{20} x_i}{20} = \frac{146}{20} = \underline{7.3 \text{ Ans.}}$

Sample variance, $s^2 = \frac{1}{(n-1)} \left\{ \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum x_i\right)^2}{n} \right\}$

$= \frac{1}{(20-1)} \left\{ 1484.76 - \frac{(146)^2}{20} \right\}$

$= \frac{1}{19} \left\{ 1484.76 - 1065.8 \right\} = \frac{418.96}{19} = 22.051$

$\therefore$ Sample Standard deviation, $s = \sqrt{s^2} = \sqrt{22.051}$

$= \underline{4.696 \text{ Ans.}}$

Coefficient of Symmetry, $\beta_1 = \frac{\mu_3}{s^3}$

$\mu_3 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^3 = \frac{1}{20} (2511.876)$

$s^3 = (4.696)^3$

$\therefore$ Coefficient of Symmetry, $\beta_1 = \frac{\frac{1}{20} (2511.876)}{(4.696)^3}$

$= \underline{1.213 \text{ Ans}}$

1(b.) MLE for $\theta = \hat{\theta} = \dfrac{n\bar{X}}{(n-1)S^2}$

For this data set,

$$\hat{\theta} = \frac{n\bar{x}}{(n-1)s^2} = \frac{(20)(7.3)}{(20-1)(22.051)}$$

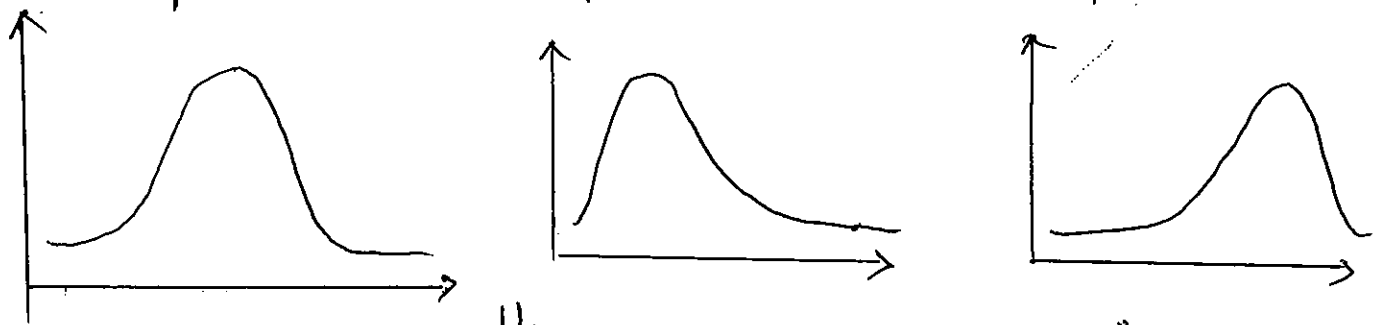
$$= \underline{0.348 \text{ Ans.}}$$

1(c) For a Gamma(a,b) distribution, the coefficient of symmetry is approximately equal to $\dfrac{2}{\sqrt{a}}$.

Since this set of data is thought to have come from Gamma$(3,\theta)$ the coef. of symmetry is approx. $\dfrac{2}{\sqrt{3}} = \underline{1.155}$ which is quite similar to the ~~value~~ 1.213 computed in 1(a).

Coefficient of symmetry is a measure of skewness.

Symmetrical	Right skewed	Left skewed.
$\beta_1 = 0$	$\beta_1 > 0$	$\beta_1 < 0$



1d) Kurtosis coef, $\gamma_2 = \dfrac{\mu_4}{S^4} - 3$. For this data it is 1.62 Since it is more than zero, the distribution is more peaked than normal curve.

2(a). Defn.

$$\text{Bias}\left[g(\underset{\sim}{X})\right] = E\left[g(\underset{\sim}{X}) - \theta\right]$$

$$\text{MSE}\left[g(\underset{\sim}{X})\right] = E\left[\left(g(\underset{\sim}{X}) - \theta\right)^2\right]$$

2(b). $X_{(n)} = \max\left(X_1, X_2, \ldots, X_n\right)$.

$$F_{X_{(n)}}(x) = P\left\{X_{(n)} \leq x\right\}.$$

$$= P\left\{\max(X_1, X_2, \ldots, X_n) \leq x\right\}.$$

$$= P\left\{X_1 \leq x, X_2 \leq x, \ldots, X_n \leq x\right\}.$$

$$= P(X_1 \leq x)\, P(X_2 \leq x) \cdots P(X_n \leq x)$$

(Since all the X_i's are independent).

$$\therefore F_{X_{(n)}}(x) = F_{X_1}(x)\, F_{X_2}(x) \cdots F_{X_n}(x)$$

$$= \left[F_{X_1}(x)\right]^n \quad \text{(Since all the } X_i\text{'s are identical)}$$

Now, $F_{X_1}(x) = \int_{-\infty}^{x} f_{X_1}(t)\, dt$

For, $x < 0$, $F_{X_1}(x) = 0$

$0 \leq x \leq \theta$, $F_{X_1}(x) = \int_0^x f_{X_1}(t)\, dt = \int_0^x \frac{2t}{\theta^2}\, dt = \frac{1}{\theta^2}\int_0^x 2t\, dt$

$$\therefore F_{X_1}(x) = \frac{1}{\theta^2}\left[t^2\right]_0^x = \frac{x^2}{\theta^2}$$

and $F_{X_1}(x) = 1$ for $x > \theta$

$$\therefore F_{X_1}(x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{\theta^2}, & 0 \le x \le \theta \\ 1, & x > \theta \end{cases}$$

Therefore,

$$F_{X_{(n)}}(x) = \left[F_{X_1}(x)\right]^n = \left[\frac{x^2}{\theta^2}\right]^n = \left[\frac{x}{\theta}\right]^{2n} \quad 0 \le x \le \theta$$

2(c). Pdf of $X_{(n)} = f_{X_{(n)}}(x) = \frac{d}{dx}\left[F_{X_{(n)}}(x)\right]$

$$= \frac{d}{dx}\left(\frac{x}{\theta}\right)^{2n}$$

$$= 2n\left(\frac{x}{\theta}\right)^{2n-1}\left(\frac{1}{\theta}\right)$$

$$\therefore f_{X_{(n)}}(x) = \frac{2n\,x^{2n-1}}{\theta^{2n}} \quad \text{for } 0 < x < \theta \text{ and zero otherwise.}$$

2(d). $E\left[X_{(n)}\right] = \int_{-\infty}^{\infty} x\, f_{X_{(n)}}(x)\,dx = \int_0^{\theta} x\left(\frac{2n\,x^{2n-1}}{\theta^{2n}}\right)dx$

$$= \frac{2n}{\theta^{2n}}\int_0^{\theta} x^{2n}\,dx = \frac{2n}{\theta^{2n}}\left[\frac{x^{2n+1}}{(2n+1)}\right]_0^{\theta}$$

$$\therefore E\left[X_{(n)}\right] = \frac{2n}{\theta^{2n}}\left[\frac{\theta^{2n+1}}{(2n+1)} - 0\right] = \frac{2n\theta}{2n+1}$$

Hence, $E\left[\frac{(2n+1)}{2n} X_{(n)}\right] = \frac{(2n+1)}{2n} E\left[X_{(n)}\right]$

$$= \frac{(2n+1)}{2n}\left(\frac{2n\theta}{2n+1}\right) = \theta$$

Since, $E\left[\frac{(2n+1)}{2n} X_{(n)}\right] = \theta$, the estimator

$\frac{(2n+1)}{2n} X_{(n)}$ is an unbiased estimator for θ.

2(e). It is given that $MSE\left(\frac{2n+1}{2n} X_{(n)}\right) = \frac{\theta^2}{4n^2(n+1)}$.

Since $\frac{(2n+1)}{2n} X_{(n)}$ is unbiased,

$$MSE\left(\frac{2n+1}{2n} X_{(n)}\right) = Var\left(\frac{2n+1}{2n} X_{(n)}\right).$$

$$= \frac{\theta^2}{4n^2(n+1)} \longrightarrow 0 \text{ as } n \longrightarrow \infty$$

Hence, it is a consistent estimator.

Remark: For definition of consistent estimator, Students may refer to "Freund, Mathematical Statistics with Applications, 8th Edition, Page 289."
See also THEOREM 10.3.

3a). $X_i \overset{iid}{\sim} \text{Beta}(\theta, 1)$, $i = 1, \ldots, n$

From the formula sheet for $\text{Beta}(\alpha_1, \alpha_2)$, the pop. mean is $\frac{\alpha_1}{\alpha_1 + \alpha_2}$.

$\therefore E(X) = \frac{\theta}{\theta + 1}$.

To obtain M.M.E of θ, we equate the pop. mean to the sample mean.

i.e., $\frac{\hat{\theta}}{\hat{\theta} + 1} = \bar{X}$

$$\hat{\theta} = \bar{X}(\hat{\theta} + 1) = \bar{X}\hat{\theta} + \bar{X}$$

$$\hat{\theta} - \bar{X}\hat{\theta} = \bar{X}$$

$$\hat{\theta}(1 - \bar{X}) = \bar{X}$$

$$\therefore \hat{\theta} = \frac{\bar{X}}{1 - \bar{X}}$$ ANS.

3b). The pdf of X_i (Beta $(\theta, 1)$) is given as,

$$f_{X_i}(x) = \begin{cases} \theta x_i^{\theta - 1}, & 0 < x_i < 1 \\ 0, & \text{otherwise.} \end{cases}$$

$$L(\theta, \underset{\sim}{x}) = \prod_{i=1}^{n} f_{x_i}(x_i)$$

$$l(\theta, \underset{\sim}{x}) = \log L(\theta, \underset{\sim}{x}) = \log \prod_{i=1}^{n} f_{x_i}(x_i)$$

$$= \sum_{i=1}^{n} \log f_{x_i}(x_i) = \sum_{i=1}^{n} \log\left(\theta x_i^{\theta-1}\right)$$

$$\therefore l(\theta, \underset{\sim}{x}) = \sum_{i=1}^{n} \left\{\log\theta + (\theta-1)\log x_i\right\}.$$

$$= \sum_{i=1}^{n} \log\theta + (\theta-1)\sum_{i=1}^{n} \log x_i$$

$$= n\log\theta + (\theta-1)\sum_{i=1}^{n} \log x_i$$

$$= n\log\theta + \theta\sum_{i=1}^{n}\log x_i - \sum_{i=1}^{n}\log x_i$$

Differentiate $l(\theta, \underset{\sim}{x})$ with respect to θ and equate it to zero.

$$\frac{d\, l(\theta, \underset{\sim}{x})}{d\theta} = \frac{n}{\theta} + \sum_{i=1}^{n} \log x_i$$

$$\therefore \frac{n}{\hat{\theta}} + \sum_{i=1}^{n} \log x_i = 0$$

$$\sum_{i=1}^{n} \log x_i = -\frac{n}{\hat{\theta}} \implies \hat{\theta} = -\frac{n}{\sum_{i=1}^{n} \log x_i}$$ ANS.

Check for maximum,

$$\frac{d^2 l(\theta, \underset{\sim}{x})}{d\theta^2} = -\frac{n}{\theta^2} < 0.$$

4a) Sort the data in ascending order.

61, 69, 71, 72, 76, 76, 77, 79,
80, 81, 83, 86, 86, 89, 90, 91, 93, 101.

$n = 19$. Median, $x_{\left(\frac{n+1}{2}\right)} = x_{\left(\frac{19+1}{2}\right)} = x_{(10)} = \underline{\underline{81}}$

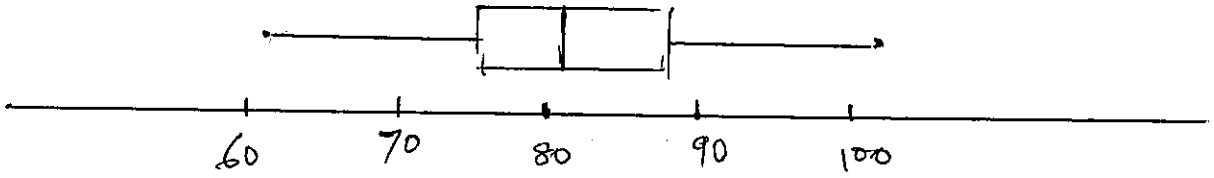
$$Q_1 = x_{(1+(n-1)0.25)} = x_{(1+(19-1)0.25)} = x_{(5.5)}$$

$$\therefore Q_1 = x_{(5.5)} = \frac{x_{(5)} + x_{(6)}}{2} = \frac{76+76}{2} = \underline{\underline{76}}$$

$$Q_3 = x_{(1+(n-1)0.75)} = x_{(1+(19-1)0.75)} = x_{(14.5)}$$

$$\therefore Q_3 = x_{(14.5)} = \frac{x_{(14)} + x_{(15)}}{2} = \frac{87+89}{2} = \underline{\underline{88}}$$

Hence, boxplot is



4b) $\frac{\bar{x}-\mu}{s/\sqrt{n}}$ is a pivotal quantity since it a function of the random sample and μ, monotonic and has a known distribution, namely $t_{(n-1)}$.

4c). A 95% C.I for μ is

$$\bar{x} \pm t_{18,0.025} \frac{s}{\sqrt{n}}$$

Given $\sum x_i = 1548$ and $\sum (x_i - \bar{x})^2 = 1670.737$.

$$\therefore \bar{x} = \frac{\sum x_i}{n} = \frac{1548}{19} = 81.474$$

$$s = \sqrt{\frac{1}{(n-1)} \left\{ \sum (x_i - \bar{x})^2 \right\}} = \sqrt{\frac{1670.737}{18}} = 9.634$$

$t_{18,0.025} = 2.101$ (From page 45, NCST – Blue book).

Hence, 95% C.I for μ is

$$81.474 \pm (2.101) \frac{9.634}{\sqrt{19}}$$

$$= 81.474 \pm 4.644$$

$$= \underline{(76.830, 86.118)} \text{ Ans.}$$

Assumption made is the data is normally distributed. which appears to be plausible ~~from the Boxplot~~.

4d) Since "85" is included in the 95% C.I obtained from (c), it appears reasonable that the population mean could be "85".

5(a). $H_0: p = 0.5$ vs. $H_1: p \neq 0.5$

Let X be the number of heads observed. $X \sim Bin(4, p)$

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

$$= P(X=0 \mid p=0.5) + P(X=4 \mid p=0.5).$$

(Since we reject H_0 if number of head is zero or 4).

$$= 0.0625 + \left(1 - P(X \leq 3 \mid p=0.5)\right).$$

From Binomial Table page 5 (NCST)

$$\therefore \alpha = 0.0625 + (1 - 0.9375).$$

$$= 0.0625 + 0.0625$$

$$\therefore \alpha = \underline{0.125 \text{ Ans.}}$$

5b) $\beta = P(\text{Type II error}) = P(\text{Do not reject } H_0 \mid H_1 \text{ is true})$.

$$= P(\text{No of heads is } 1, 2 \text{ or } 3 \mid p = 0.7)$$

$$= 1 - P(X=0 \mid p=0.7) - P(X=4 \mid p=0.7)$$

$$\therefore \beta = 1 - \left[\binom{4}{0}(0.7)^0(0.3)^4\right] - \left[\binom{4}{4}(0.7)^4(0.3)^0\right]$$

$$= 1 - \left[(1)(1)(0.0081)\right] - \left[(1)(0.2401)(1)\right].$$

$$= 1 - 0.0081 - 0.2401$$

$$\therefore \beta = \underline{0.7518 \text{ Ans.}}$$

6 a) $n = 10$

$\sum x_i = 56.8$, $\sum x_i^2 = 356.4$, $\sum x_i y_i = 329.364$

$\sum y_i = 52.74$, $\sum y_i^2 = 306.786$.

$$S_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n} = 356.4 - \frac{(56.8)^2}{10} = 33.776$$

$$S_{yy} = \sum y_i^2 - \frac{(\sum y_i)^2}{n} = 306.786 - \frac{(52.74)^2}{10} = 28.635$$

$$S_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n} = 329.364 - \frac{(56.8)(52.74)}{10}$$

$$= 29.801$$

Therefore, sample correlation coefficient,

$$r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} = \frac{29.801}{\sqrt{(33.776)(28.635)}} = \frac{29.801}{31.1}$$

$\therefore$ $r = \underline{0.9582}$ Ans.

Since r is quite close to "1" there is a very strong positive linear association.

6b). $H_0: \rho = 0.9$

$H_1: \rho > 0.9$

$$W = \frac{1}{2}\log\left(\frac{1+r}{1-r}\right) \sim N\left(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\right).$$

Under H_0; $\rho = 0.9$ $\sim N\left(\frac{1}{2}\log\frac{(1\cdot9)}{(0\cdot1)}, \frac{1}{10-3}\right)$.

$$W \sim N(1.4722, 0.143).$$

Test statistic

$$Z = \frac{\frac{1}{2}\log\left(\frac{1+0.9582}{1-0.9582}\right) - 1.4722}{\sqrt{0.143}}.$$

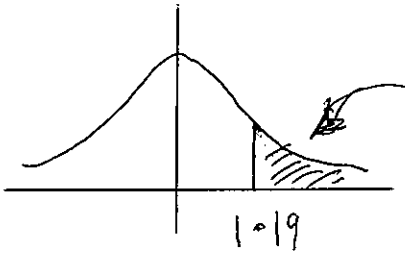
$$= \frac{\frac{1}{2}\log 46.8469 - 1.4722}{\sqrt{0.143}}$$

$$= \frac{0.4512}{0.3782} = 1.19$$

p-value $= P(Z > 1.19) = 1 - P(Z \leq 1.19)$

$= 1 - 0.8830$

$= 0.117$



Taking $\alpha = 0.05$, since p-value $= 0.117 > 0.05$,

We do not have sufficient evidence to reject H_0.